RAJALAKSHMI ENGINEERING COLLEGE

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C2

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|---|---|
| **Started on** | Thursday, 20 August 2026, 10:26 AM |
| **State** | Finished |
| **Completed on** | Thursday, 20 August 2026, 10:32 AM |
| **Time taken** | 6 mins 17 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

**Problem Statement**

Given an array of 1s and 0s this has all 1s first followed by all 0s. Aim is to find the number of 0s. Write a program using Divide and Conquer to Count the number of zeroes in the given array.

Input Format

First Line Contains Integer m – Size of array

Next m lines Contains m numbers – Elements of an array

Output Format

First Line Contains Integer – Number of zeroes present in the given array.

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>
#include <stdlib.h>

// Function to find the index of the first 0 using Divide and Conquer (Binary Search)
int findFirstZero(int arr[], int low, int high) {
    if (high >= low) {
        // Calculate mid point to avoid overflow
        int mid = low + (high - low) / 2;

        // Check if mid element is the first 0
        if ((mid == 0 || arr[mid - 1] == 1) && arr[mid] == 0) {
            return mid;
        }

        // If the element at mid is 1, the first 0 must be in the right half
        if (arr[mid] == 1) {
            return findFirstZero(arr, mid + 1, high);
        }

        // Otherwise, the first 0 is in the left half
        return findFirstZero(arr, low, mid - 1);
    }

    // If there are no 0s in the array
    return -1;
}

int main() {
    int m;

    // Read the size of the array
    if (scanf("%d", &m) != 1) {
        return 0;
    }

    int *arr = (int *)malloc(m * sizeof(int));
    if (arr == NULL) {
        return 1;
    }

    // Read the elements of the array
    for (int i = 0; i < m; i++) {
        if (scanf("%d", &arr[i]) != 1) {
            free(arr);
            return 0;
        }
    }

    // Find the first index where 0 appears
    int firstZeroIndex = findFirstZero(arr, 0, m - 1);

    // Calculate total number of zeroes
```

```
52 |    // Calculate total number of zeroes
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| ✔ | 5<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |
| ✔ | 10<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1 | 0 | 0 | ✔ |
| ✔ | 8<br>0<br>0<br>0<br>0<br>0<br>0<br>0<br>0 | 8 | 8 | ✔ |
| ✔ | 17<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>1<br>0<br>0 | 2 | 2 | ✔ |

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

Back to Course